INCH-POUND
MIL-M-38510/321C
14 July 2003
SUPERSEDING
MIL-M-38510/321B
12 August 1983

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, BUFFERS/DRIVERS, OPEN COLLECTOR OUTPUT, HIGH VOLTAGE, MONOLITHIC SILICON

Inactive for new design after 18 April 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, buffers/drivers microcircuits with open collector high voltage outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part number. The part number should be in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types should be as follows:

Device type	Circuit
02	Quad 2-input inverter buffer/driver, 15-volt output

1.2.2 Device class. The device class should be the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines should be as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P. O. Box 3990, Columbus, OH 43216-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation, (P_D) (per device) 1/	+24.2 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}): Cases A, B, C, D, and 2	(See MIL-STD-1835)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	+4.5 V dc minimum to +5.5 V dc maximum
Minimum high level input voltage (V_{IH})	+2.0 V dc
Maximum low level input voltage (V_{IL})	+0.7 V dc
Normalized fanout (each output) 2/	10 maximum
Case operating temperature range (T_C)	-55° to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Departments of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-883 - Test Method Standard for Microelectronics
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.3.2 Truth tables. The truth tables shall be as specified on figure 2.

3.3.3 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity upon request.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-PRF-38535, appendix A).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ 1/ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	2/	02		15	V
Input clamp voltage	V_{IC}	V_{CC} = +4.5 V, I_{IN} = -18 mA, T_C = +25°C	02		-1.5	V
Maximum collector cut-off current	I_{CEX1}	V_{CC} = +4.5 V, V_{OH} = +12 V	02		50	µA
Maximum collector cut-off current	I_{CEX2}	V_{CC} = +4.5 V, V_{OH} = +15 V	02		250	µA
Low level output voltage	V_{OL}	V_{CC} = +4.5 V, I_{OL} = +4 mA,	02		0.4	V
High level input current	I_{IH1}	V_{CC} = +5.5 V, V_{IN} = +2.7 V 3/	02		20	µA
High level input current	I_{IH2}	V_{CC} = +5.5 V, V_{IN} = +5.5 V 3/	02		100	µA
Low level input current	I_{IL}	V_{CC} = +5.5 V, V_{IN} = +0.4 V 4/	02	-.03	-.40	mA
High level supply current	I_{CCH}	V_{CC} = +5.5 V, V_{IN} = 0 V	02		1.6	mA
Low level supply current	I_{CCL}	V_{CC} = +5.5 V, V_{IN} = 5.5 V	02		4.4	mA
Propagation delay time high to low level	t_{PHL}	C_L = 50 pF ±10% R_L = 2 kΩ ±5%	02	2	46	ns
Propagation delay time low to high level	t_{PLH}		02	2	52	ns

1/ Complete terminal conditions shall be as specified in table III.
2/ See I_{CEX}.
3/ All unspecified inputs grounded.
4/ All unspecified inputs at 5.5 volts.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance Inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 9	1*, 2, 3, 9
Group A test requirements	1, 2, 3, 9, 10, 11	1, 2, 3, 9,
Group B electrical test parameters when using the method 5005 QCI option	1, 2, 3, 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 9, 10, 11	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	10. 11
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

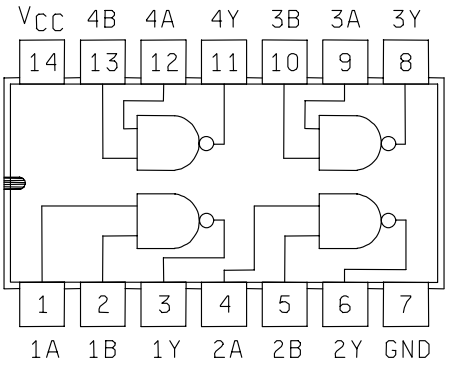
4.5 Methods of inspection. Methods of inspection shall be specified and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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Device type 02

CASES A,B,C, AND D



CASES 2

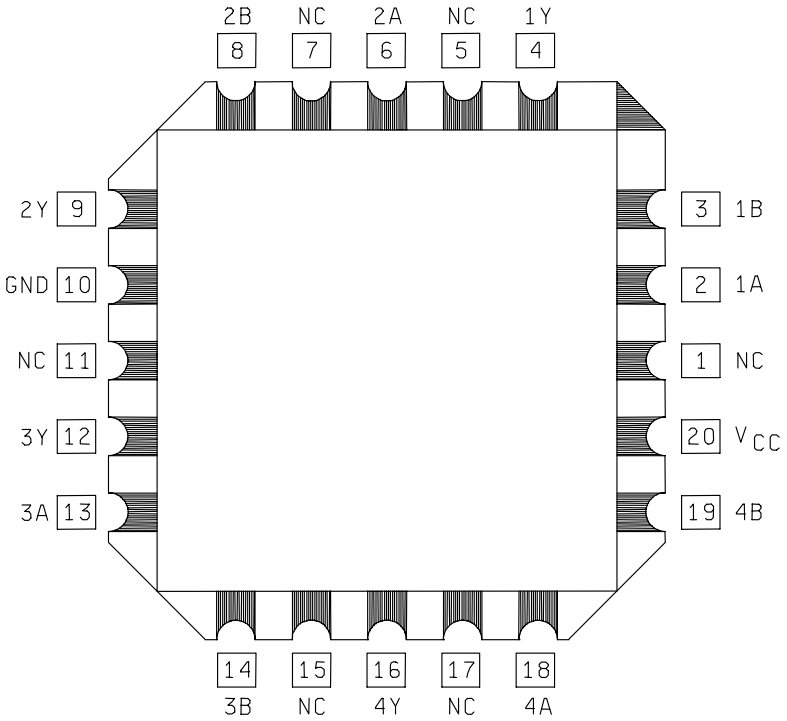


FIGURE 1. Terminal connections and logic diagrams.

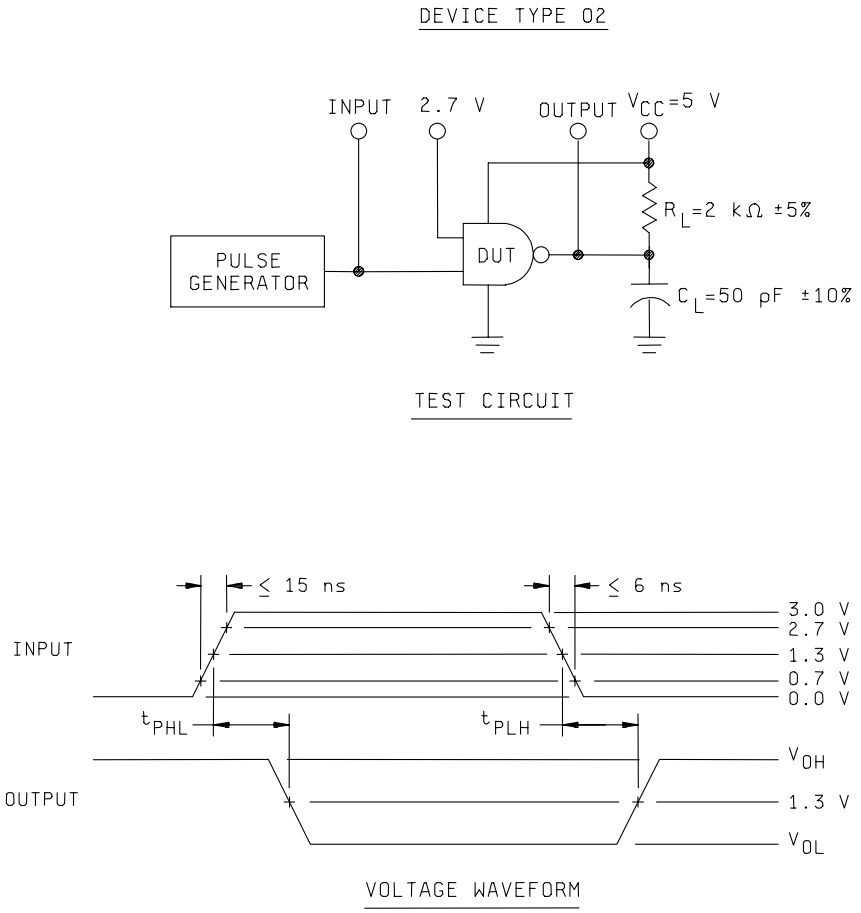
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Device type 02

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic Y = $\overline{AB}$

FIGURE 2. Truth table.



NOTES:
1. Pulse generator has the following characteristics: PRR ≤ 1 MHz, and $Z_{OUT} \approx 50\Omega$.
2. C_L = 50 pF ±10% including scope probe, wiring and stray capacitance without package in test fixture.
3. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 3. Switching time test circuit.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 3.0 V, or low ≤ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OL}	3007	1	2.0 V	2.0 V	4 mA				GND							4.5 V	1Y		0.4	V
		"	2				2.0 V	2.0 V	4 mA	"							"	2Y		"	"
		"	3							"	4 mA	2.0 V	2.0 V				"	3Y		"	"
		"	4							"				4 mA	2.0 V	2.0 V	"	4Y		"	"
	I_{CEX1}		5	0.7 V	2.0 V	12 V				"							"	1Y		50	µA
			6	2.0 V	0.7 V	12 V				"							"	1Y		"	"
			7				0.7 V	2.0 V	12 V	"							"	2Y		"	"
			8				2.0 V	0.7 V	12 V	"							"	2Y		"	"
			9							"	12 V	0.7 V	2.0 V				"	3Y		"	"
			10							"	12 V	2.0 V	0.7 V				"	3Y		"	"
			11							"				12 V	0.7 V	2.0 V	"	4Y		"	"
			12							"				12 V	2.0 V	0.7 V	"	4Y		"	"
	I_{CEX2}		13	0.7 V	2.0 V	15 V				"							"	1Y		250	"
			14	2.0 V	0.7 V	15 V				"							"	1Y		"	"
			15				0.7 V	2.0 V	15 V	"							"	2Y		"	"
			16				2.0 V	0.7 V	15 V	"							"	2Y		"	"
			17							"	15 V	0.7 V	2.0 V				"	3Y		"	"
			18							"	15 V	2.0 V	0.7 V				"	3Y		"	"
			19							"				15 V	0.7 V	2.0 V	"	4Y		"	"
			20							"				15 V	2.0 V	0.7 V	"	4Y		"	"
	I_{IH1}	3010	21	2.7 V	GND					"							5.5 V	1A		20	"
		"	22	GND	2.7 V					"							"	1B		"	"
		"	23				2.7 V	GND		"							"	2A		"	"
		"	24				GND	2.7 V		"							"	2B		"	"
		"	25							"		2.7 V	GND				"	3A		"	"
		"	26							"		GND	2.7 V				"	3B		"	"
		"	27							"					2.7 V	GND	"	4A		"	"
		"	28							"					GND	2.7 V	"	4B		"	"
	I_{IH2}	"	29	5.5 V	GND					"							"	1A		100	"
		"	30	GND	5.5 V					"							"	1B		"	"
		"	31				5.5 V	GND		"							"	2A		"	"
		"	32				GND	5.5 V		"							"	2B		"	"
		"	33							"		5.5 V	GND				"	3A		"	"
		"	34							"		GND	5.5 V				"	3B		"	"
		"	35							"					5.5 V	GND	"	4A		"	"
		"	36							"					GND	5.5 V	"	4B		"	"
	I_{IL}	3009	37	0.4 V	5.5 V					"							"	1A	2/	2/	mA
		"	38	5.5 V	0.4 V					"							"	1B	"	"	"
		"	39				0.4 V	5.5 V		"							"	2A	"	"	"
		"	40				5.5 V	0.4 V		"							"	2B	"	"	"
		"	41							"		0.4 V	5.5 V				"	3A	"	"	"
		"	42							"		5.5 V	0.4 V				"	3B	"	"	"
		"	43							"					0.4 V	5.5 V	"	4A	"	"	"
		"	44							"					5.5 V	0.4 V	"	4B	"	"	"
	I_{CCL}	3005	45	5.5 V	5.5 V		5.5 V	5.5 V		"		5.5 V	5.5 V		5.5 V	5.5 V	"	V_{CC}		4.4	"
	I_{CCH}	3005	46	GND	GND		GND	GND		"		GND	GND		GND	GND	"	V_{CC}		1.6	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 3.0 V, or low ≤ 0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2 1/	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}		Min	Max	
1 Tc = 25°C	V_{IC}		47	-18 mA						GND							4.5 V	1A		-1.5	V
			48		-18 mA					"							"	1B		"	"
			49				-18 mA			"							"	2A		"	"
			50					-18 mA		"							"	2B		"	"
			51							"		-18 mA					"	3A		"	"
			52							"			-18 mA				"	3B		"	"
			53							"					-18 mA		"	4A		"	"
			54							"						-18 mA	"	4B		"	"
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
9 Tc = 25°C	t_{PHL}	3003 Fig. 3	55	IN	2.7 V	OUT				GND							5.0 V	1A to 1Y	2	33	ns
		"	56				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	57							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	58							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
	t_{PLH}	"	59	IN	2.7 V	OUT				"							"	1A to 1Y	"	40	"
		"	60				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	61							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	62							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
10 Tc = 125°C	t_{PHL}	"	63	IN	2.7 V	OUT				"							"	1A to 1Y	"	46	"
		"	64				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	65							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	66							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
	t_{PLH}	"	67	IN	2.7 V	OUT				"							"	1A to 1Y	"	52	"
		"	68				IN	2.7 V	OUT	"							"	2A to 2Y	"	"	"
		"	69							"	OUT	IN	2.7 V				"	3A to 3Y	"	"	"
		"	70							"				OUT	IN	2.7 V	"	4A to 4Y	"	"	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

1/ Case 2 pins not referenced are NC.

2/ I_{IL} limits in mA are as follows:

Min/Max limits for circuit		
A	B	C
-.160/-.400	-.030/-.300	-.150/-.380

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Complete part number (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

j. Requirements for "JAN" marking.

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

GND ..... Ground zero voltage potential
I_{IN} ..... Current flowing into an input terminal
V_{IN} ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
02	54LS26

6.8 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Manufacturer		
	Texas Instruments	Signetics Corp.	Motorola Inc.
02 circuit	A	B	C

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-1968)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

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3. DOCUMENT TITLE
MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, BUFFERS/DRIVERS, OPEN COLLECTOR OUTPUT, MONOLITHIC SILICON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus	b. TELEPHONE *(Include Area Code* (1) Commercial 614-692-0536 (2) DSN 850-0536
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